

Car Evaluation Dataset

Data information

Derived from a simple hierarchical decision model, this dataset may be useful for te

- **buying**: vhigh, high, med, low.
- **maint**: vhigh, high, med, low.
- **doors**: 2, 3, 4, 5.
- **persons**: 2, 4, 5.
- **lug_boot**: small, med, big.
- **safety**: low, med, high.
- **class**: unacc, acc, good, vgood

Enter Google Drive link of the CSV file:

https://drive.google.com/file/d/1w3e5vRk1xZDHs0bLgBE6R70PSH43BUCH/view?usp=share_link

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 1727 entries, 0 to 1726
```

```
Data columns (total 7 columns):
```

#	Column	Non-Null Count	Dtype
0	vhigh	1727 non-null	object
1	vhigh.1	1727 non-null	object
2	2	1727 non-null	object
3	2.1	1727 non-null	object
4	small	1727 non-null	object
5	low	1727 non-null	object
6	unacc	1727 non-null	object

```
dtypes: object(7)
```

```
memory usage: 94.6+ KB
```

Analysis:

Missing Data: There are no missing values in any of the columns, as indicated by

Data Types: The categorical variables are represented as object type.

Memory Usage: The dataset uses 94.6 KB of memory, which is manageable for most a

Overall: The dataset is clean with no missing values, and the data types are rel

Data size and dimension

Dataset shape: (1727, 7)

	buying	maint	doors	persons	lug_boot	safety	class
0	vhigh	vhigh	2	2	small	med	unacc
1	vhigh	vhigh	2	2	small	high	unacc
2	vhigh	vhigh	2	2	med	low	unacc
3	vhigh	vhigh	2	2	med	med	unacc
4	vhigh	vhigh	2	2	med	high	unacc

This dataset consists of 1727 rows and 7 columns. Each row represents a unique car e

- **buying**: Buying price of the car (Categorical)
- **maint**: Maintenance price of the car (Categorical)
- **doors**: Number of doors in the car (Numerical)
- **persons**: Capacity in terms of persons to carry (Numerical)
- **lug_boot**: Size of the luggage boot (Categorical)
- **safety**: Estimated safety of the car (Categorical)
- **class**: Evaluation level of the car (Target variable, Categorical)

Missing Data

Missing Data Counts:

	0
buying	0
maint	0
doors	0
persons	0
lug_boot	0
safety	0
class	0



- The analysis of missing data in the dataset reveals that there are no missing values.
- The heatmap visualization further supports this finding, as it does not display any white or light-colored cells.

Duplicated Values

No duplicated rows found.

Number of Duplicated Rows: 0

- The analysis of duplicate data in the dataset indicates that there are no duplicated rows.

Numerical data columns

Summary statistics of the numerical columns:

	doors	persons
--	-------	---------

count	1,727	1,727
mean	3.5009	3.6676
std	1.1181	1.2473
min	2	2
25%	3	2
50%	4	4
75%	4.5	5
max	5	5

Analysis:

- **Numeric Columns**: `doors` and `persons` are the two numeric columns in the data
- **Unique Values**: At the beginning, the unique values in these two columns `doors`
- **Central Tendency**: The mean and median values for both `doors` (at 3.5) and `pe
- **Dispersion**: The standard deviation values indicate the spread of the data arou
- **Range**: The range (difference between the minimum and maximum values) for both
- **Percentiles**: The percentiles provide insights into the distribution of the dat
- **Outliers**: The minimum and maximum values for both columns are within a reasona

Categorical data columns

Categorical Data Columns:

0
buying
maint
lug_boot
safety
class

Unique Values in Column 'buying':

value
high

vhigh
high
med
low

buying	count
high	432
med	432
low	432
vhigh	431

Unique Values in Column 'maint':

value
vhigh
high
med
low

maint	count
high	432
med	432
low	432
vhigh	431

Unique Values in Column 'lug_boot':

value
small
med

big

lug_boot	count
med	576
big	576
small	575

Unique Values in Column 'safety':

value

med

high

low

safety	count
med	576
high	576
low	575

Unique Values in Column 'class':

value

unacc

acc

vgood

good

class	count
unacc	1,209
acc	384
vgood	213
good	134

good	69
vgood	65

Unique Values:

- buying: ['vhigh', 'high', 'med', 'low']
- maint: ['vhigh', 'high', 'med', 'low']
- lug_boot: ['small', 'med', 'big']
- afety: ['low', 'med', 'high']
- class: ['unacc', 'acc', 'vgood', 'good']

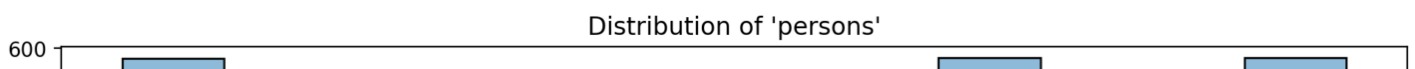
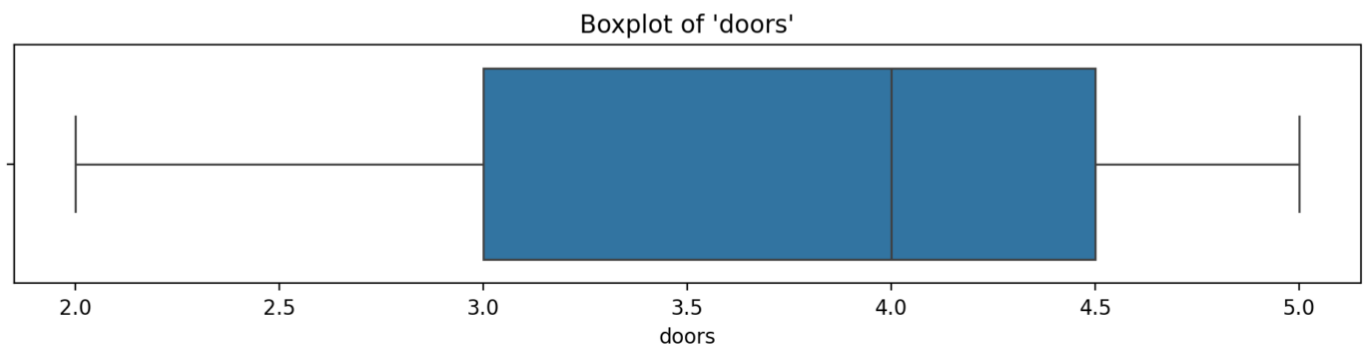
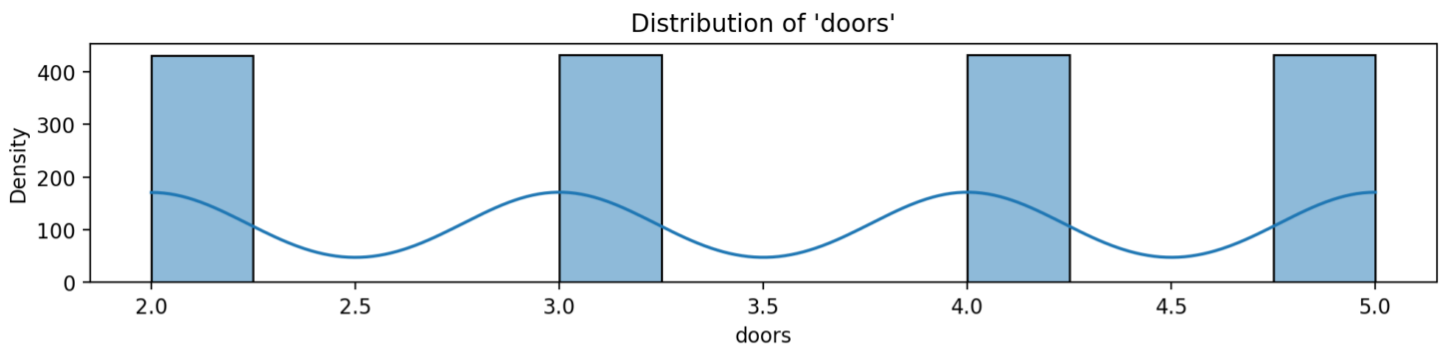
Analysis:

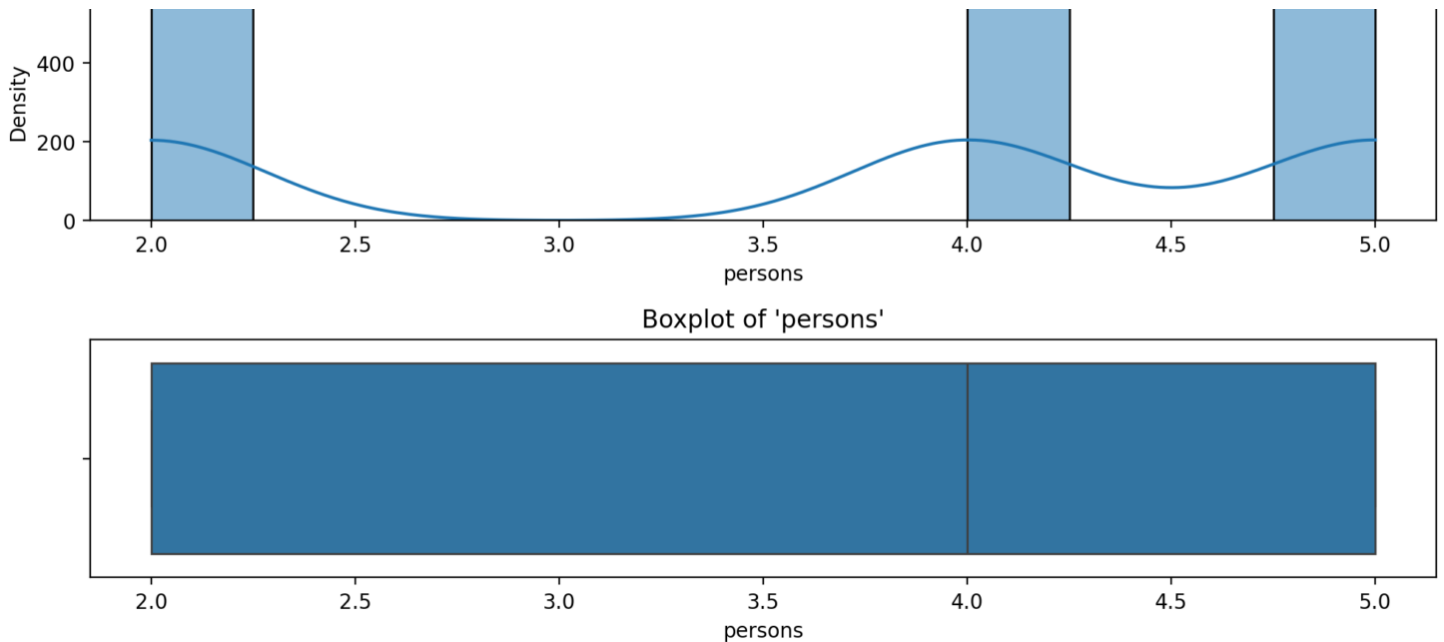
- In `buying`, the values are distributed relatively evenly with each unique value a
- In `maint`, approximately 432 occurrences of each unique value indicate a general
- In `lug\_boot`, each unique value appears around 576 times, indicating a fairly equ
- In `safety`, the values are distributed relatively evenly with each unique value a
- In `class`, the values are distributed unevenly, with "unacc" accounting for 1210

Overall:

- The data of categorical values does not look unusual. Each unique value has comple
- Almost the categorical data in this dataset is evenly distributed. However, the va

Plotting Distribution and Box Plots for Numeric Columns





****Number of Columns**:**

- There are ****2 numeric columns**** in the dataset: ``doors`` and ``persons``.

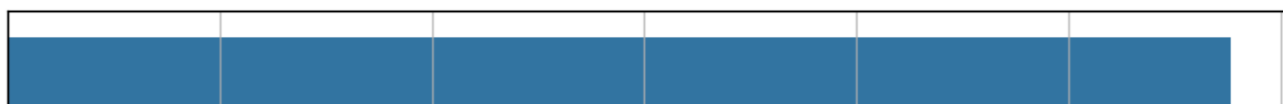
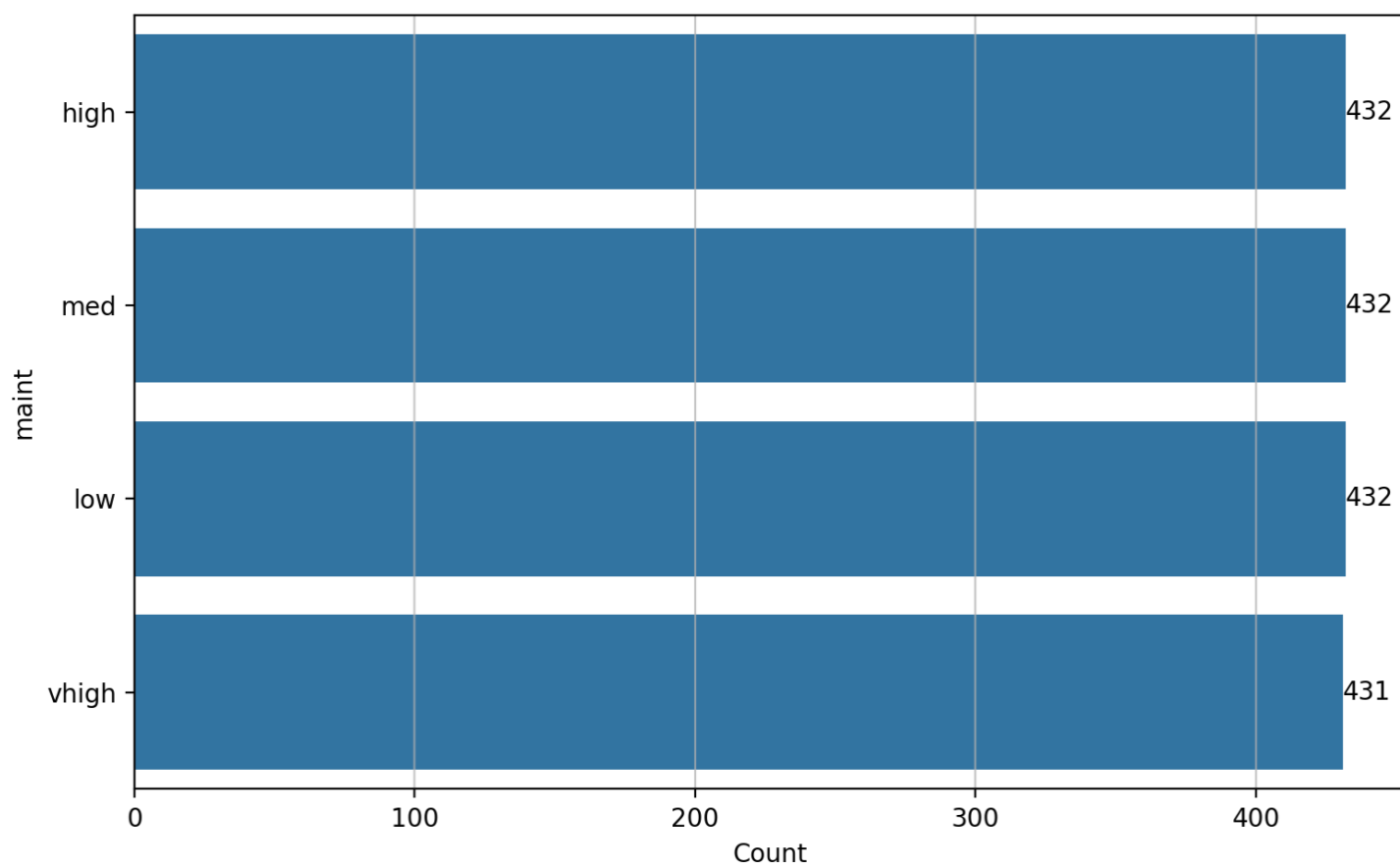
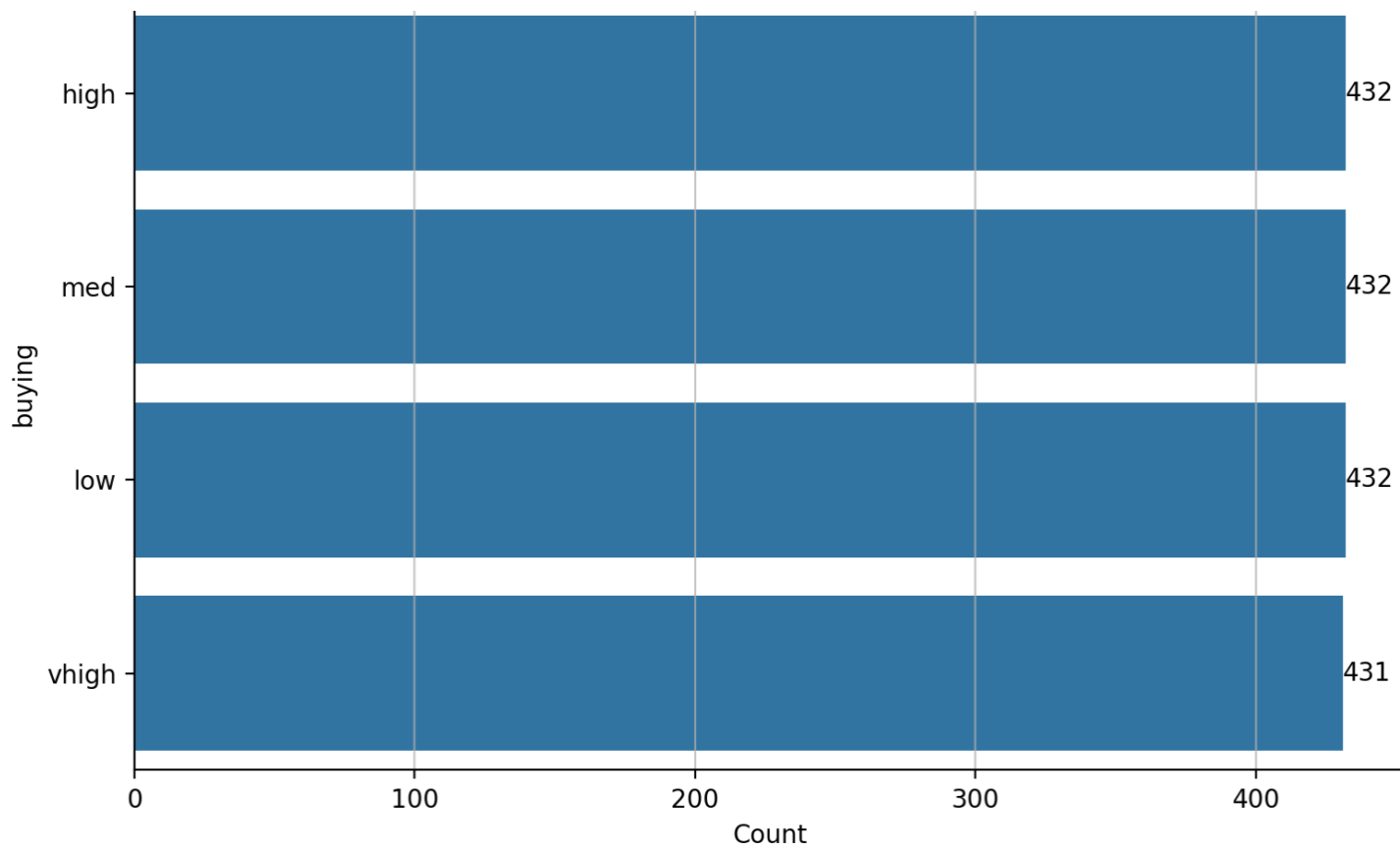
****Shapes Drawn for Each Column**:**

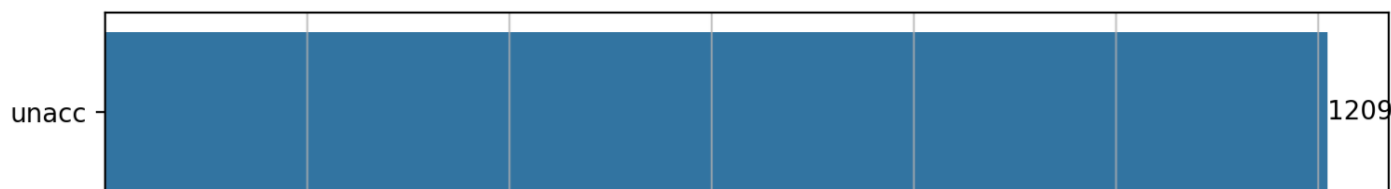
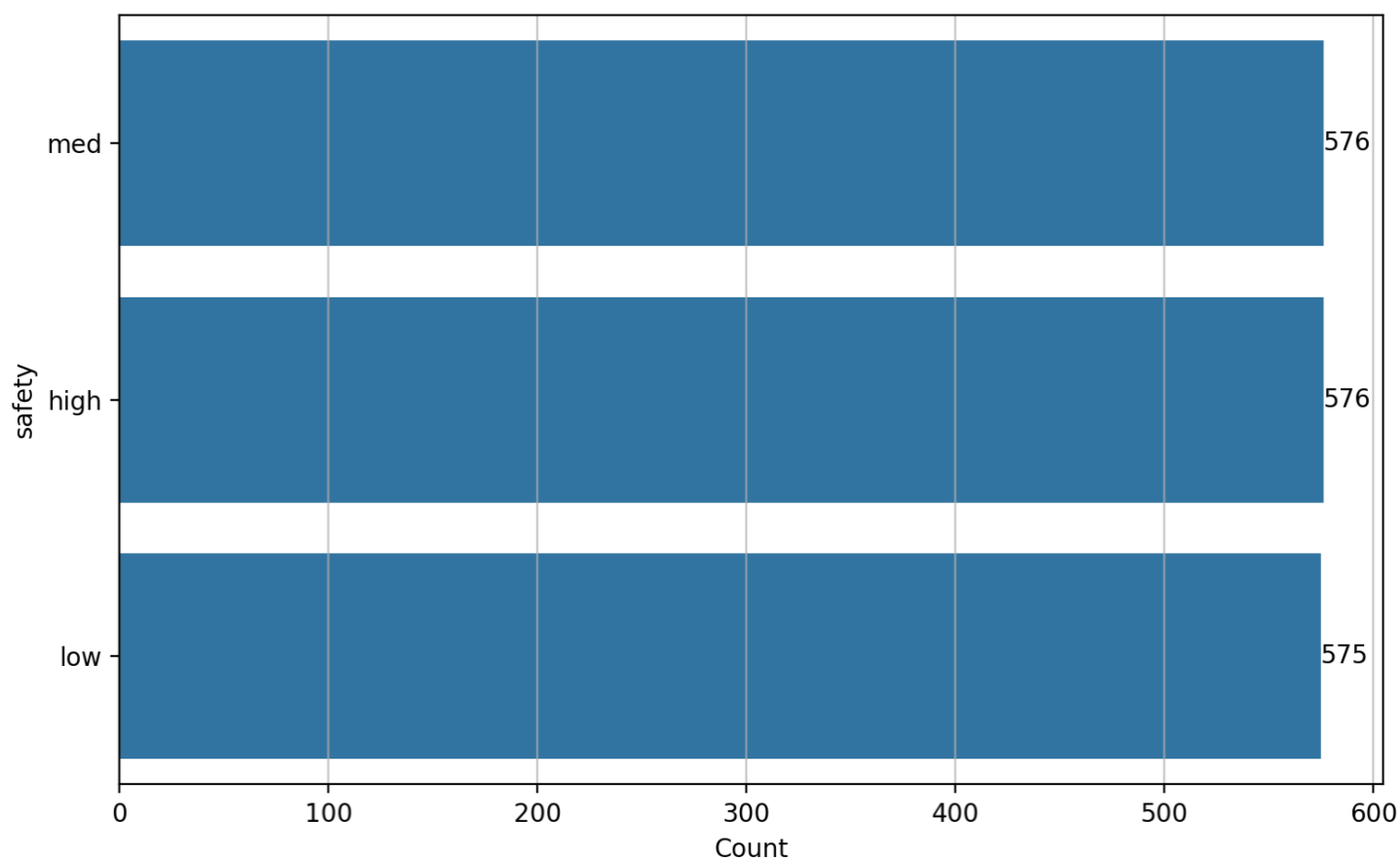
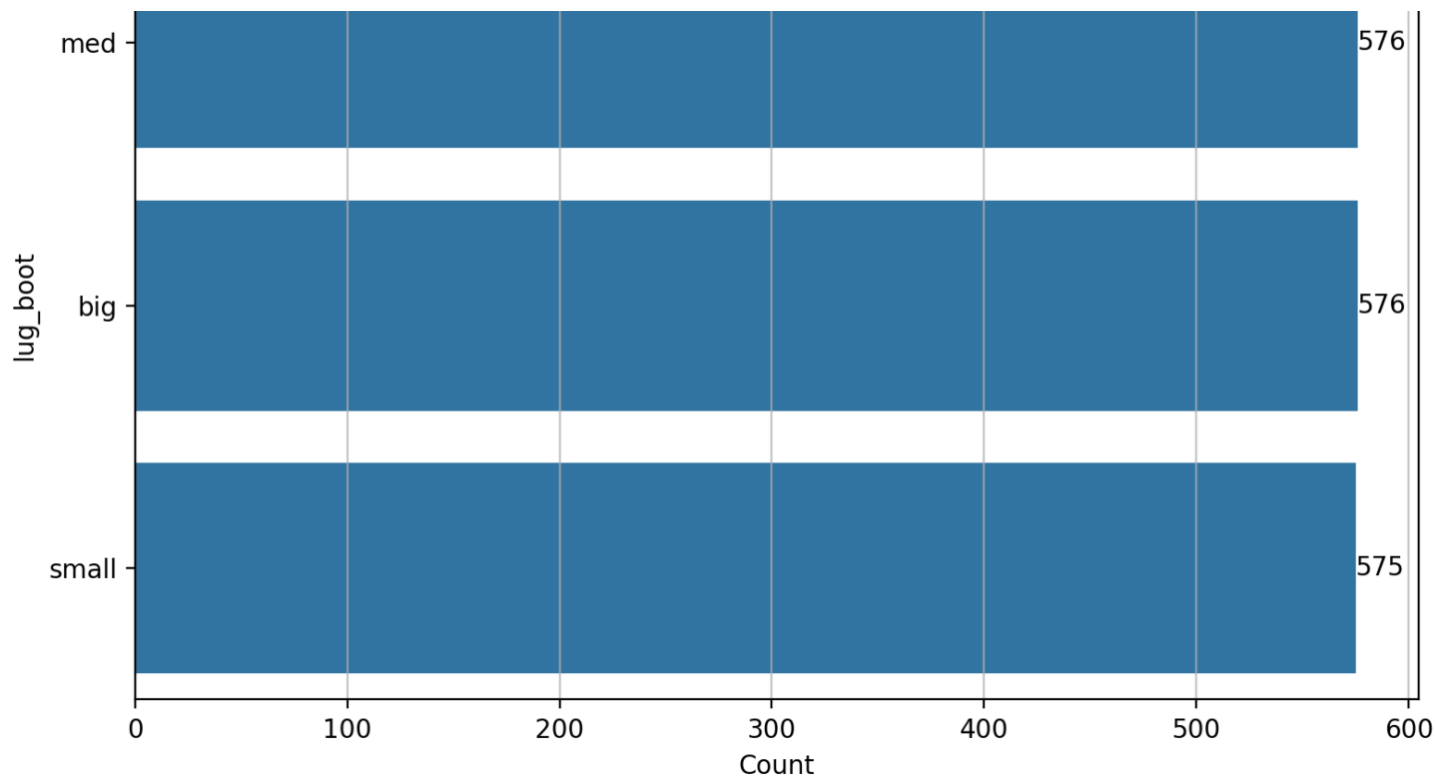
- For each numeric column, two types of plots are drawn:
 1. ****Distribution Plot (Histogram with KDE)****: This plot shows the distribution
 2. ****Box Plot****: This plot provides a summary of the data distribution, highlighting

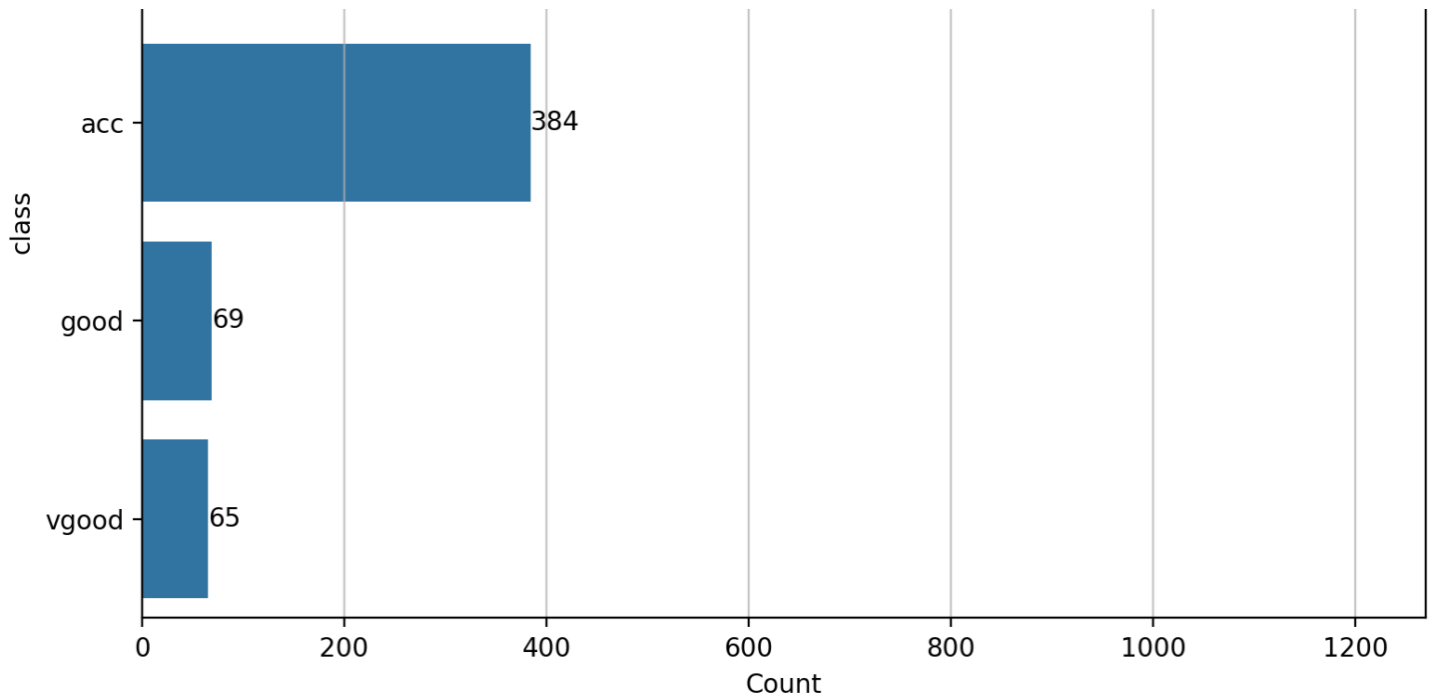
****Analysis of Distribution and Histogram Shapes**:**

- ****`doors`****:
 - ****Data values****: [``2.0``, ``3.0``, ``4.0``, ``5.0``]
 - ****Distribution Plot****: The histogram shows the frequency distribution of the
 - ****Box Plot****: The median value (the line inside the box) is at 3.5, indicating
 - ****Overall****: The values of ``doors`` are distributed uniformly and according to
- ****`persons`****:
 - ****Data values****: [``2.0``, ``4.0``, ``5.0``]
 - ****Distribution Plot****: The histogram shows the frequency distribution of the
 - ****Box Plot****: The median value (the line inside the box) is at 4.0, indicating
 - ****Overall****: The values of ``persons`` are not distributed uniformly because of

Plotting Horizontal Bar Plots for Categorical Columns







****Number of Columns**:**

- There are **5 categorical values** in the dataset: `buying`, `maint`, `lug\_boot`

****Shapes Drawn for Each Column**:**

- For each categorical column, there is a **Distribution Plot** are drawn, which

****Analysis of Distribution and Histogram Shapes**:**

- **`buying`, `maint`**:
 - **Distribution Plot**: There are 4 columns in the histogram for `buying` and
 - In `buying` and `maint`, the values are distributed almost evenly (the frequ
- **`lug\_boot`**:
 - **Distribution Plot**: There are 3 columns in the histogram for `lug\_boot` sh
 - In `lug\_boot`, the values are distributed almost evenly (the frequent of eac
- **`safety`**:
 - **Distribution Plot**: There are 3 columns in the histogram for `safety` show
 - In `safety`, the values are distributed almost evenly (the frequent of each
- **`class`**:
 - **Distribution Plot**: There are 4 columns in the histogram for `class` shows
 - In `class`, the values are distributed unevenly (the frequent `unacc` is cou

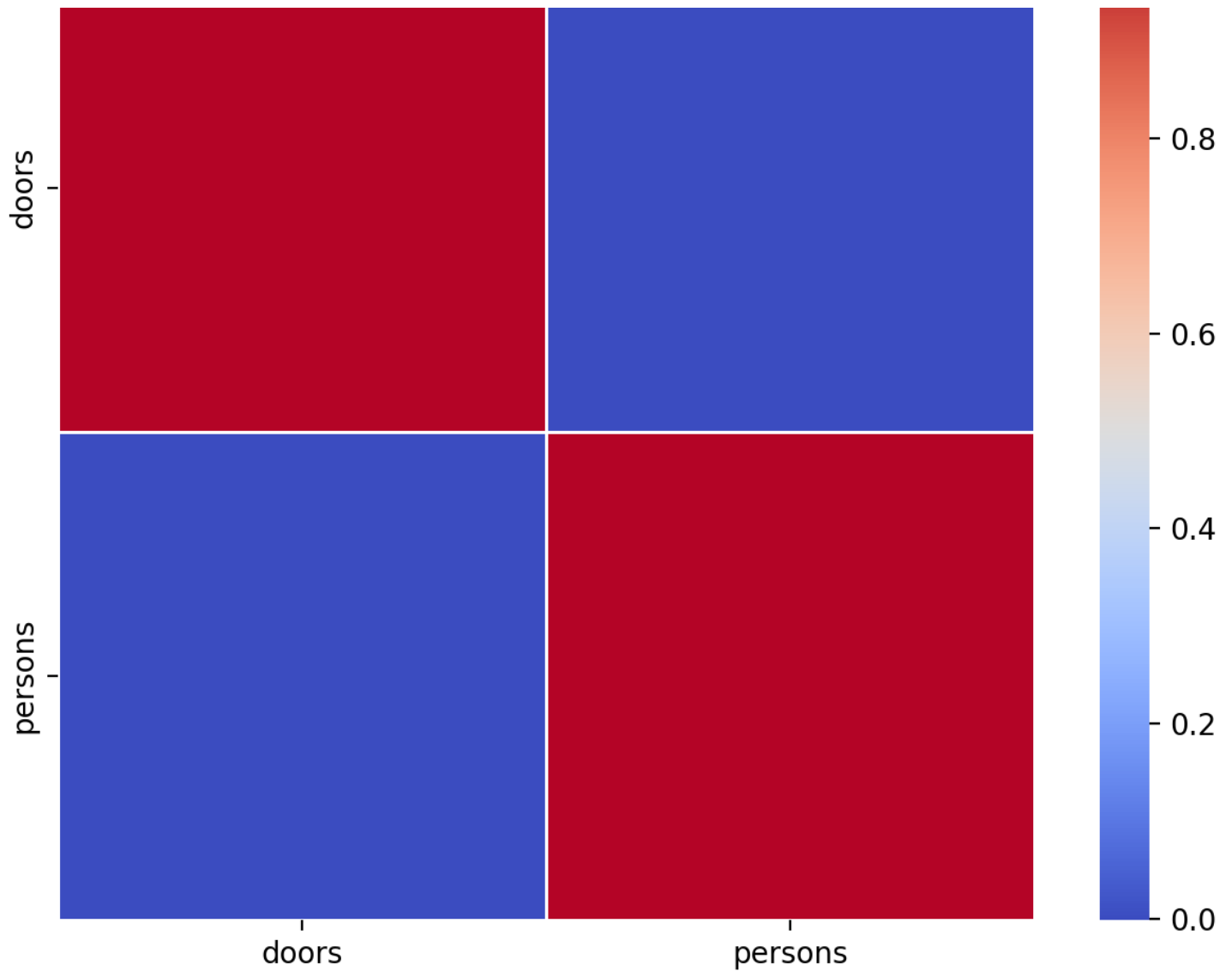
****Overall****

- In general, except for the uneven distribution of `class` data, the remaining c

Correlation Matrix

Heatmap of Correlation Matrix





Purpose of the Heatmap:

- The heatmap visualizes the correlation matrix for five numeric columns: 'doors' and 'persons'.

Analysis of Correlation Patterns:

- Diagonal Elements: The diagonal elements (top-left and bottom-right) are red, indicating a correlation of 1.0.
- Off-Diagonal Elements: The off-diagonal elements (top-right and bottom-left) are blue, indicating a correlation of 0.0.
- Relevance: This heatmap is relevant for understanding the relationships between the variables.